

Airline Fleet Revenue Analysis: Identifying Occupancy Gaps and Profit Opportunities

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1. Executive summary

This report provides a data-driven analysis of fleet performance and revenue optimization using a sqlite database. The study identifies the primary financial drivers and operational inefficiencies within the current booking model.

Key Findings:

- **Revenue Distribution** - The Economy class is the core volume driver, generating ₱ 14,695,684,400 (70.76% of total revenue), while the Business class provides high-margin growth with an average ticket price of ₱ 51,143.42.
- **Operational Anomaly** - A significant discrepancy was identified between the SU9 and 773 aircraft models. Despite the 773 having 402 seats and a 0.66 occupancy rate, the SU9 generates significantly higher total revenue (₱ 5.11 billion vs. ₱ 3.43 billion) due to its high-frequency utilization of 6,931 flights.
- **Growth Opportunity** - A "What-If" analysis confirms that a 10% increase in occupancy across the fleet is a transformative opportunity. This strategic shift could elevate the SU9 turnover to over ₱ 5.63 billion and the 763 to ₱ 4.81 billion without increasing fixed operational costs.

2. Technical Foundation

2.1 About the Data

The analysis is based on an aviation sqlite database containing comprehensive information across eight interconnected tables, including `aircrafts_data`, `airport_data`, `flights`, `ticket_flights`, `boarding_passes`, `bookings`, `seats` and `tickets`. The dataset captures a high-volume booking window from end-June 2017 to mid-August 2017. Key data attributes include flight schedules, seat configurations, and tiered pricing

structures across various aircraft models. All financial figures are presented in original database units without currency conversion.

The dataset specifically never stated which currency is used in this data, but the names of the aircraft model have Russian in it so we can assume that this airplane dataset is from Russia and the currency used here is Russian Ruble (₽).

2.2 Methodology

To extract actionable insights from the raw database, the following technical approach was implemented:

- Data Extraction & Cleaning - Integrated Python with the sqlite database to perform complex relational joins. Data was audited for completeness, confirming zero null values in the primary financial columns.
- Time-Series Processing - Applied datetime transformations to analyze fluctuations in daily ticket bookings and identify peak travel windows.
- Occupancy Modeling - Calculated aircraft efficiency by comparing actual seats_booked against the num_seats defined in the fleet manifest.
- Simulation - Developed a "What-If" revenue model to project the financial impact of a 10% increase in seat utilization across the entire fleet.

3. Problem statement & objectives

3.1 The problem

In a highly competitive aviation market, simply selling tickets is not enough. To maintain a competitive edge, the airline must understand which fare classes are most profitable, which aircraft are

underperforming in terms of seat occupancy, and how daily booking fluctuations affect revenue. Without this data-driven approach, the airline risks inefficient fleet allocation and lost revenue from empty seats.

3.2 Project objectives

- **Analyze Revenue & Pricing Architecture** - Evaluate the financial contribution of different fare classes (Economy, Business, Comfort) and determine the average ticket price across various aircraft models to identify high-margin segments.
- **Evaluate Operational Velocity** - Measure fleet utilization by analyzing the relationship between total flight frequency, seat capacity, and cumulative turnover to identify the most productive assets.
- **Assess Fleet Efficiency** - Calculate average occupancy rates for each aircraft model to pinpoint underutilized assets and "empty-seat" costs.
- **Quantify Growth Opportunities** - Conduct a "What-If" analysis to project the total turnover increase resulting from a strategic 10% improvement in seat occupancy.

4. Revenue analysis by fare class

This section evaluates the financial contribution of each travel tier across the entire fleet. By aggregating ticket sales from the `ticket_flights` table, we can identify which fare conditions act as the primary volume drivers and which provide the highest profit margins.

The data tells us that the Economy class is the undisputed revenue leader, generating ₱14,695,684,400, representing 70.76% of total turnover. While the Business class accounts for a smaller portion of total tickets, its premium pricing structure results in an average ticket price of ₱51,143.42,

contributing a substantial 26.51% to the bottom line. The Comfort class remains a specialized segment, contributing 2.73% of total revenue. This distribution demonstrates a classic "volume vs. value" strategy, where the high-volume Economy segment provides a stable financial baseline while the high-margin Business segment optimizes profitability per seat.

	fare_conditions	ticket_count	total_revenue	avg_ticket_price	revenue_share_pct
1	Economy	920793	14695684400	15959.81	70.76
2	Business	107642	5505179600	51143.42	26.51
3	Comfort	17291	566116900	32740.55	2.73

Table 1 - Revenue and average ticket price by fare

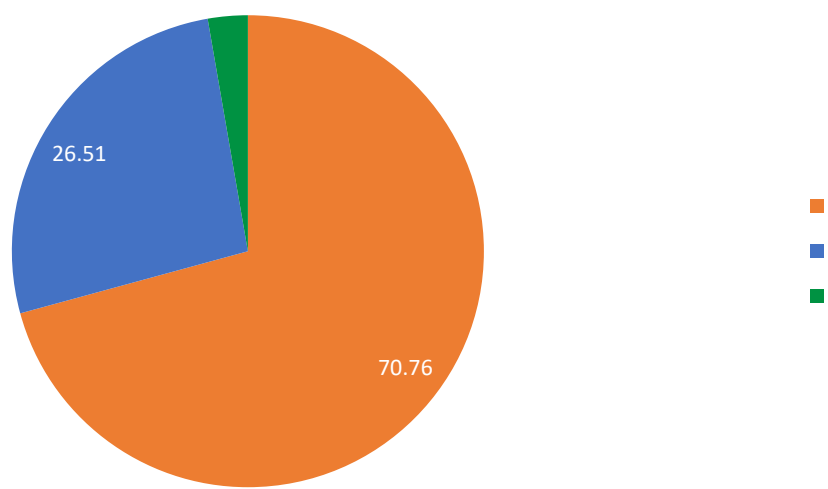


Figure 1- Distribution of revenue across fare classes

5. Fleet performance & daily trends

Our analysis compared different aircraft code (such as the 773, 763, and 319) to see which ones are the most productive.

5.1 Aircraft revenue contribution

To understand the revenue-generating potential of each model, we analyzed the average ticket price across both aircraft types and fare conditions. This reveals how the airline positions its assets, specifically showing that some models are dedicated to premium, high-yield routes while others serve as high-volume carriers.

The data shows significant variation in pricing strategies; for example, the 319 and 763 models command significantly higher average ticket prices in Business class compared to the rest of the fleet. Conversely, the SU9 and CR1 models maintain a much lower average economy class ticket price. And the 773 is the only aircraft which has comfort class. On the other side both CN1 and CR2 only have economy class. By mapping these price points against fare conditions, we can see that the airline's bottom line is protected by a balanced mix of high-volume Economy sales and high-margin Business premiums across specific strategic models.

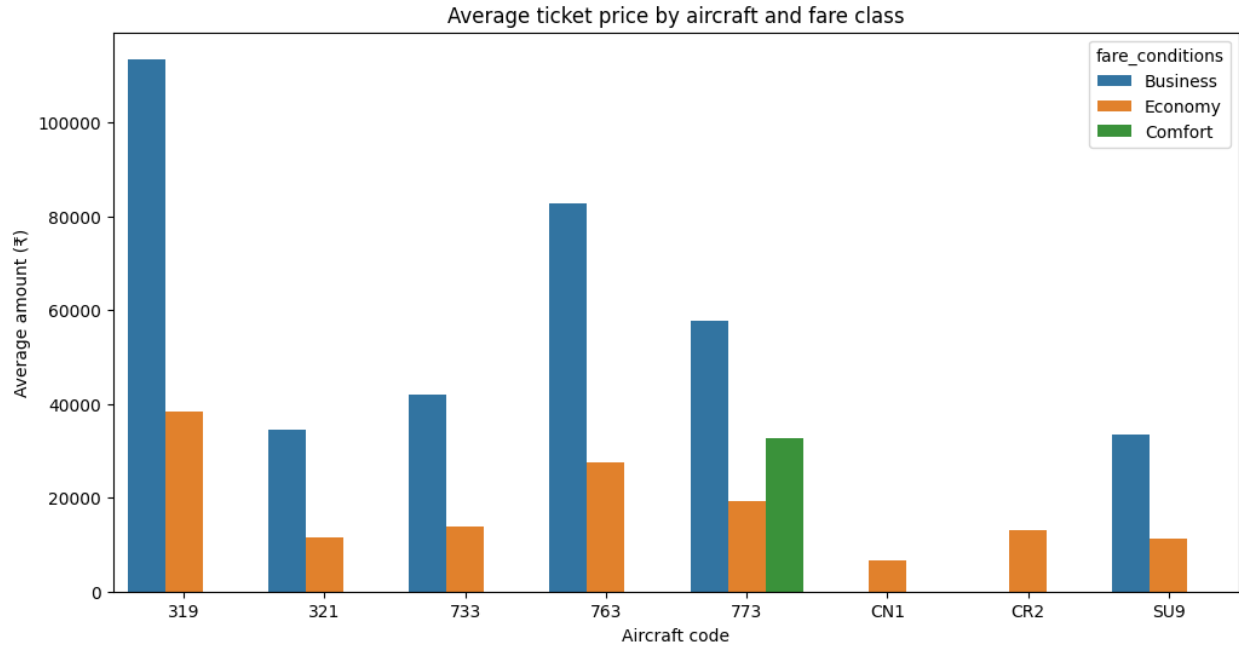


Figure 2 - Average price of ticket per aircraft model and fare class

5.2 Booking fluctuations:

Bookings ramped sharply from near-zero in late June to a sustained plateau of approximately 7,500–8,000 daily tickets by early July, suggesting the dataset captures the opening of a booking window or the start of a peak travel season. The sharp spike at the end of the period warrants further investigation into whether it reflects a data anomaly or a specific promotional event or special occasions.

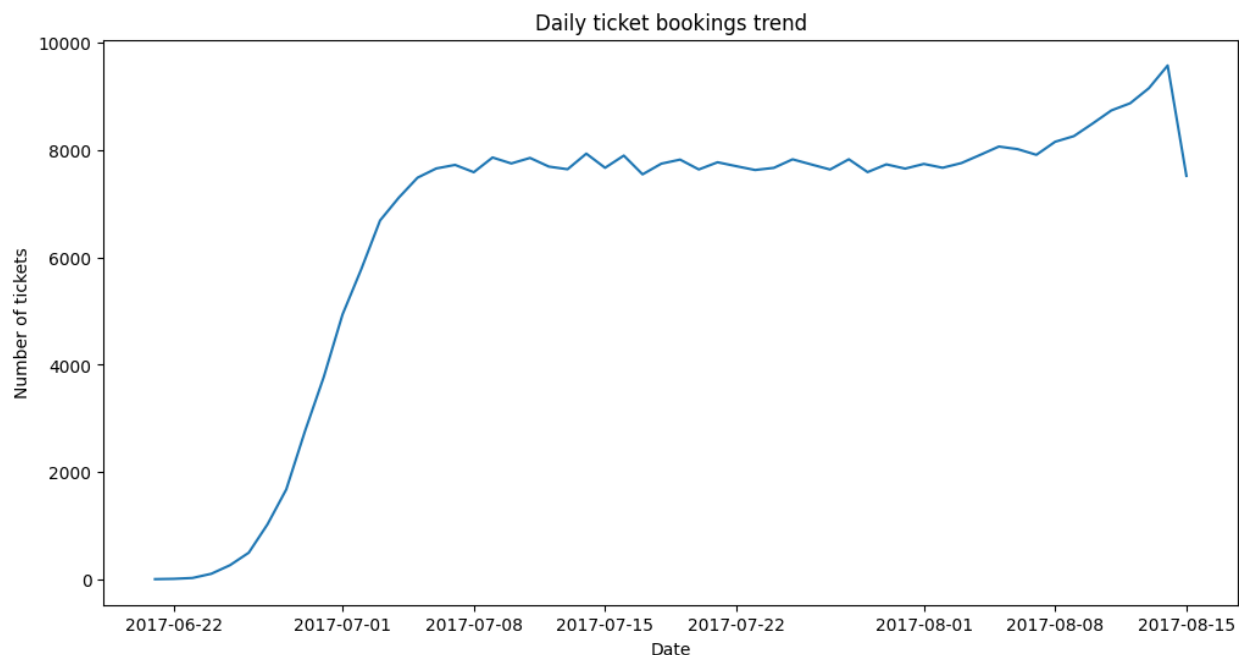


Figure 3 - Daily booking trends over time

6. Occupancy rate:

The Occupancy Rate serves as a Key Performance Indicator (KPI) for fleet efficiency, measuring the ratio of seats filled against the total available capacity for each aircraft model. Low occupancy rates indicate "perishable inventory", seats that represent a fixed operational cost (fuel, crew, and maintenance) but generate no revenue for a specific flight leg.

6.1 Fleet efficiency insights:

The data reveals varying degrees of efficiency across the fleet, with occupancy rates ranging from 0.43 to 0.66. The 773 leads the fleet in efficiency with a 66% occupancy rate, suggesting strong demand for its high-capacity configuration. Conversely, the CR2 and 319 models show the lowest utilization at 43% and 46% respectively. These underutilized models represent the most significant opportunity for revenue recovery, since the fixed costs of these flights are already incurred, any strategic improvement in filling these empty seats translates directly into high-margin turnover.

aircraft_code	seats_booked	num_seats	occupancy_rate
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1	319	53.58	116	0.46
2	321	88.81	170	0.52
3	733	80.26	130	0.62
4	763	113.94	222	0.51
5	773	264.93	402	0.66
6	CN1	6.00	12	0.50
7	CR2	21.48	50	0.43
8	SU9	56.81	97	0.59

Table 2- Average occupancy rate per aircraft model

6.2 Current revenue

Revenue table reveals a stark contrast between fleet capacity and financial output. While the 773 aircraft model features the largest capacity in the fleet with 402 seats and maintains a strong 0.66 occupancy rate, its total revenue of ₱3,431,205,500 is surprisingly outperformed by smaller models. Specifically, the SU9, despite having only 97 seats and a lower 0.59 occupancy rate, projects a significantly higher revenue of ₱5,114,484,700. This suggests that the SU9 may be deployed on higher-frequency routes or premium-priced segments that allow it to generate more total revenue than the high-capacity 773.

A similar performance trend is observed with the 763, which generates ₱4,371,277,100 in revenue from a 222-seat configuration and a 0.51 occupancy rate. These metrics raise important questions regarding internal operations and fleet deployment: the data indicates that physical seat volume and occupancy percentages are not the sole drivers of total turnover. Instead, route-specific pricing and flight frequency appear to play a more dominant role in the airline's overall revenue distribution across these different aircraft models.

aircraft_code	total_revenue
---------------	---------------

0	319	2706163100
1	321	1638164100
2	733	1426552100
3	763	4371277100
4	773	3431205500
5	CN1	96373800
6	CR2	1982760500
7	SU9	5114484700

Table 3 – Total revenue per aircraft

6.3 Fleet Utilization & Flight Frequency

The analysis of fleet utilization reveals that operational frequency is the primary driver of total turnover, effectively explaining why smaller aircraft can outperform larger models in gross revenue. While the 773 possesses the highest revenue per flight (5,634,163) due to its massive 402-seat capacity, it completed only 609 flights during the study period, resulting in a total revenue of 3,431,205,500. In stark contrast, the SU9 functions as the high-velocity of the fleet, despite having only 97 seats and a much lower 737,914 revenue per flight, it completed a staggering 6,931 flights. This high frequency allowed the SU9 to aggregate a total revenue of 5,114,484,700, significantly surpassing the high-capacity 773.

These findings highlight that physical seat volume and occupancy rates are only part of the financial equation. Assets like the CR2 and SU9 demonstrate that maintaining a high volume of departures can compensate for lower per-flight margins and smaller aircraft footprints. For internal operations, this suggests that maximizing the number of flights for efficient, high-demand models is a more effective strategy for driving total turnover than relying solely on the massive capacity of underutilized long-haul aircraft like the 773.

	aircraft_code	total_flights	total_tickets_sold	total_revenue	avg_ticket_price	revenue_per_flight
0	CR2	7424	150122	1982760500	13207.66	267074.0
1	SU9	6931	365698	5114484700	13985.54	737914.0
2	CN1	2542	14672	96373800	6568.55	37912.0
3	321	1294	107129	1638164100	15291.51	1265969.0
4	763	1195	124774	4371277100	35033.56	3657972.0
5	733	1164	86102	1426552100	16568.16	1225560.0
6	319	1067	52853	2706163100	51201.69	2536235.0
7	773	609	144376	3431205500	23765.76	5634163.0

Table 4 - Aircraft performance analysis

7. "What if" analysis: the 10% growth scenario

This section quantifies the financial opportunity hidden within current operational gaps. By applying a hypothetical 10% increase to the existing occupancy rates of each aircraft model, we can project the potential surge in Total Turnover without the need for additional flights or increased overhead costs.

7.1 Financial Projection & Logic:

The simulation reveals that even a modest improvement in seat utilization yields massive financial gains due to the high volume of existing operations.

- High-Impact Models - The SU9 shows the most dramatic potential, with a projected Expected Turnover of 5,625,933,170 following a 10% occupancy boost.
- Capacity Optimization - Larger aircraft like the 763 and 773 would see their turnovers climb to 4,808,404,810 and 3,774,326,050, respectively.

- The Bottom Line - Because the fixed costs of these flights (fuel, crew, and airport fees) are already committed, the revenue generated from this 10% increase in passengers represents nearly pure profit for the airline. This proves that focusing on filling "empty air" is the most efficient path to scaling total revenue.

	aircraft_code	seats_booked	num_seats	occupancy_rate	exp_occupancy_rate	exp_turnover
1	319	53.58	116	0.46	0.51	2976779410.00
2	321	88.81	170	0.52	0.57	1801980510.00
3	733	80.26	130	0.62	0.68	1569207310.00
4	763	113.94	222	0.51	0.56	4808404810.00
5	773	264.93	402	0.66	0.72	3774326050.00
6	CN1	6.00	12	0.50	0.55	106011180.00
7	CR2	21.48	50	0.43	0.47	2181036550.00
8	SU9	56.81	97	0.59	0.64	5625933170.00

Table 5 - Potential revenue increase (10% improved occupancy)

8. Strategic recommendations

Based on the quantitative analysis of fleet utilization, fare structures, and occupancy metrics, the following data-driven strategies are recommended to optimize total turnover:

- Prioritize High-Velocity Assets: The data proves that the SU9 generates higher total revenue (5.11 billion) than the larger 773 (3.43 billion) due to its superior flight frequency. The airline should prioritize the maintenance and scheduling of high-frequency "workhorse" models for short-haul, high-demand corridors to maximize cumulative turnover.

- **Targeted Occupancy Recovery for Low-Yield Models:** Aircraft like the CR2 and 319 currently operate at the lowest efficiencies, with occupancy rates of 43% and 46% respectively. Implementing dynamic pricing or "last-minute" promotional offers specifically for these models could capture the "perishable inventory" and shift these flights toward the more profitable 0.60+ occupancy range.
- **Re-evaluate Long-Haul Deployment:** The 773 has the highest revenue per flight (P 5,634,163) but the lowest frequency. To increase its total revenue contribution, the airline should explore increasing its departure frequency or deploying it on higher-demand routes where its 402-seat capacity can be fully leveraged alongside its lead-leading 0.66 occupancy rate.
- **Capitalize on Business Class Margins:** Since Business Class generates 26.51% of total revenue from a relatively small number of tickets (107,642), marketing efforts should focus on upselling programs. Even a small conversion of Economy passengers to Business, especially on high-price models like the 319 where the Business average is 113,535, would significantly boost total turnover without increasing passenger volume.
- **Adopt the 10% Occupancy KPI:** The "What-If" analysis confirms that a 10% increase in seat utilization is a transformative revenue driver. This 10% growth target should be adopted as a formal Key Performance Indicator (KPI) for the next fiscal quarter to turn underutilized capacity into earned revenue.

9. Conclusion

The analysis concludes that the airline's financial success is driven more by operational velocity (flight frequency) than by aircraft scale (seat capacity). The high-frequency deployment of the SU9 fleet proves that consistent asset utilization is the most effective way to aggregate high turnover. However, the

lower occupancy rates observed in models like the CR2 (43%) and 319 (46%) indicate substantial "perishable inventory" that remains unmonetized.

By implementing the recommended 10% occupancy growth strategy, the airline can capitalize on fixed-cost flights to significantly boost total turnover, potentially reaching an expected turnover of P 5.63 billion for high-velocity models like the SU9. Transitioning from a volume-heavy focus to a utilization-centric model will allow the airline to maximize the revenue potential of every seat, turning underutilized capacity into a sustainable competitive advantage